

1. Write a Python program to filter a list of integers using Lambda.

Original list of integers:

```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

Even numbers from the said list:

```
[2, 4, 6, 8, 10]
```

Odd numbers from the said list:

```
[1, 3, 5, 7, 9]
```

2. Write a Python program to sort a list of tuples using Lambda.

Original list of tuples:

```
[('English', 88), ('Science', 90), ('Maths', 97), ('Social sciences', 82)]
```

Sorting the List of Tuples:

```
[('Social sciences', 82), ('English', 88), ('Science', 90), ('Maths', 97)]
```

3. Write a Python program to square and cube every number in a given list of integers using Lambda.

Original list of integers:

```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

Square every number of the said list:

```
[1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
```

Cube every number of the said list:

```
[1, 8, 27, 64, 125, 216, 343, 512, 729, 1000]
```

4. Write a Python program to find if a given string starts with a given character using Lambda.

Sample Output:

```
True
```

```
False
```

5. Write a Python program to check whether a given string is a number or not using Lambda.

6. Write a Python program to count the even and odd numbers in a given array of integers using Lambda.

Original arrays:

[1, 2, 3, 5, 7, 8, 9, 10]

Number of even numbers in the above array: 3

Number of odd numbers in the above array: 5

7. Write a Python program to add two given lists using map and lambda.

Original list:

[1, 2, 3]

[4, 5, 6]

Result: after adding two list

[5, 7, 9]

8. Write a Python program to find palindromes in a given list of strings using Lambda.

Original list of strings:

['php', 'w3r', 'Python', 'abcd', 'Java', 'aaa']

List of palindromes:

['php', 'aaa']

9. Write a Python program to remove None values from a given list using the lambda function.

Original list:

[12, 0, None, 23, None, -55, 234, 89, None, 0, 6, -12]

Remove None value from the said list:

[12, 0, 23, -55, 234, 89, 0, 6, -12]

10. Write a Python program to calculate the sum of the positive and negative numbers of a given list of numbers using the lambda function.

Original list: [2, 4, -6, -9, 11, -12, 14, -5, 17]

Sum of the positive numbers: -32

Sum of the negative numbers: 48